

SLIDES

# Lesson 1 — The workbook as a test lab

AI agents on the workbook

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# Contents

- 1 Introduction
  - 1.1 The workbench as a test lab
  - 1.2 The AI agent can:
  - 1.3 Let the machine do the boring work, and keep the overview...
- 2 Workflow
  - 2.1 Ishikawa Circle
  - 2.2 The board
  - 2.3 The test cycle
  - 2.4 A test in Python
  - 2.5 Baud rate and time
  - 2.6 Manual or with an agent?
  - 2.7 Setup
  - 2.8 Thank you!

# The workbook as a test lab

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AI agents that flash, measure and test on their own



# THE AI AGENT CAN:

- Flash firmware
- Watch the serial log
- Run tests
- Validate the results
- Fix problems in the code

→ Your workbench becomes an automated test lab

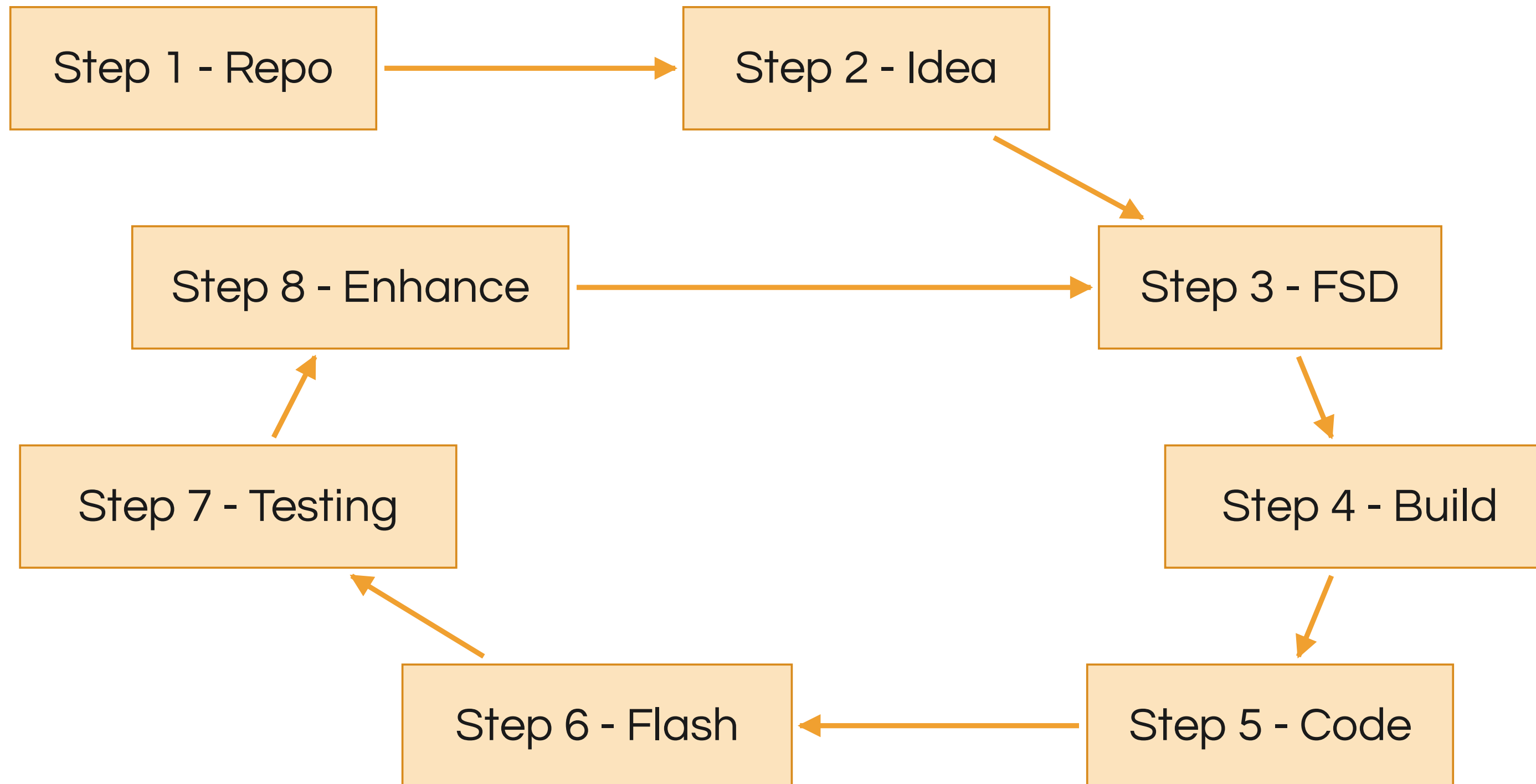


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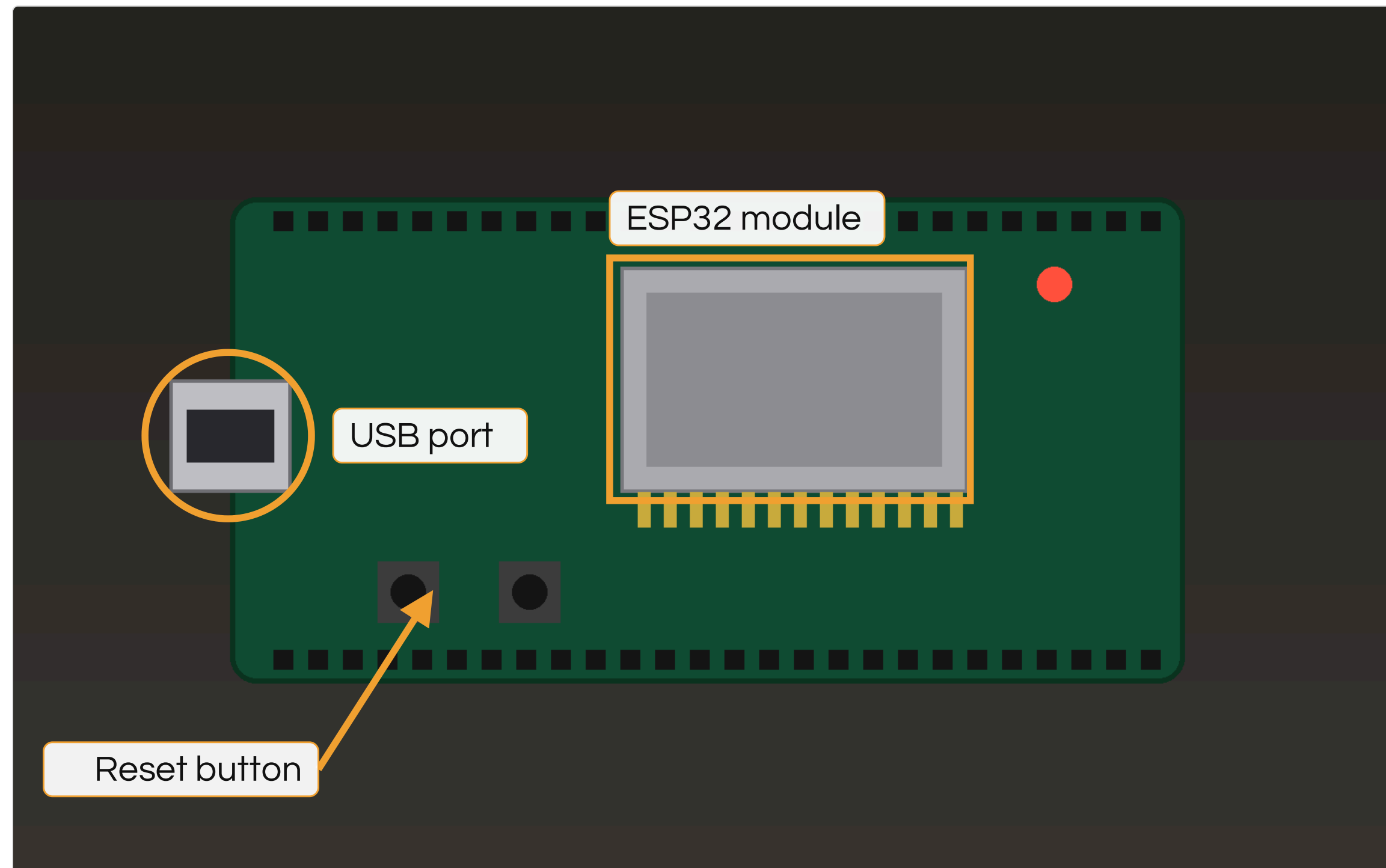
Let the machine do the boring work, and  
keep the overview yourself.



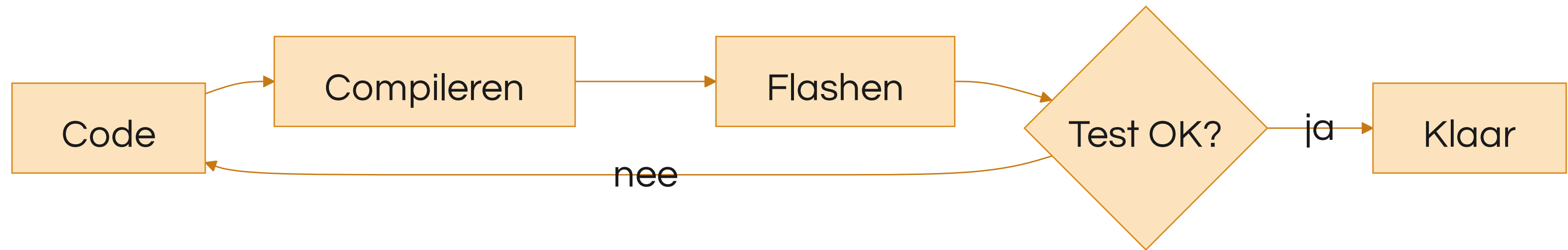
# Ishikawa Circle



# THE BOARD



# THE TEST CYCLE



# A TEST IN PYTHON

```
import serial
port = serial.Serial("/dev/ttyUSB0", 115200, timeout=10)

def wait_for(text):
    line = port.read_until(text.encode())
    return text in line.decode()

assert wait_for("BOOT OK")
```



# BAUD RATE AND TIME

$$t_{byte} = \frac{10}{\text{baud rate}}$$

At 115200 baud, one byte takes about  $87 \mu s$ .



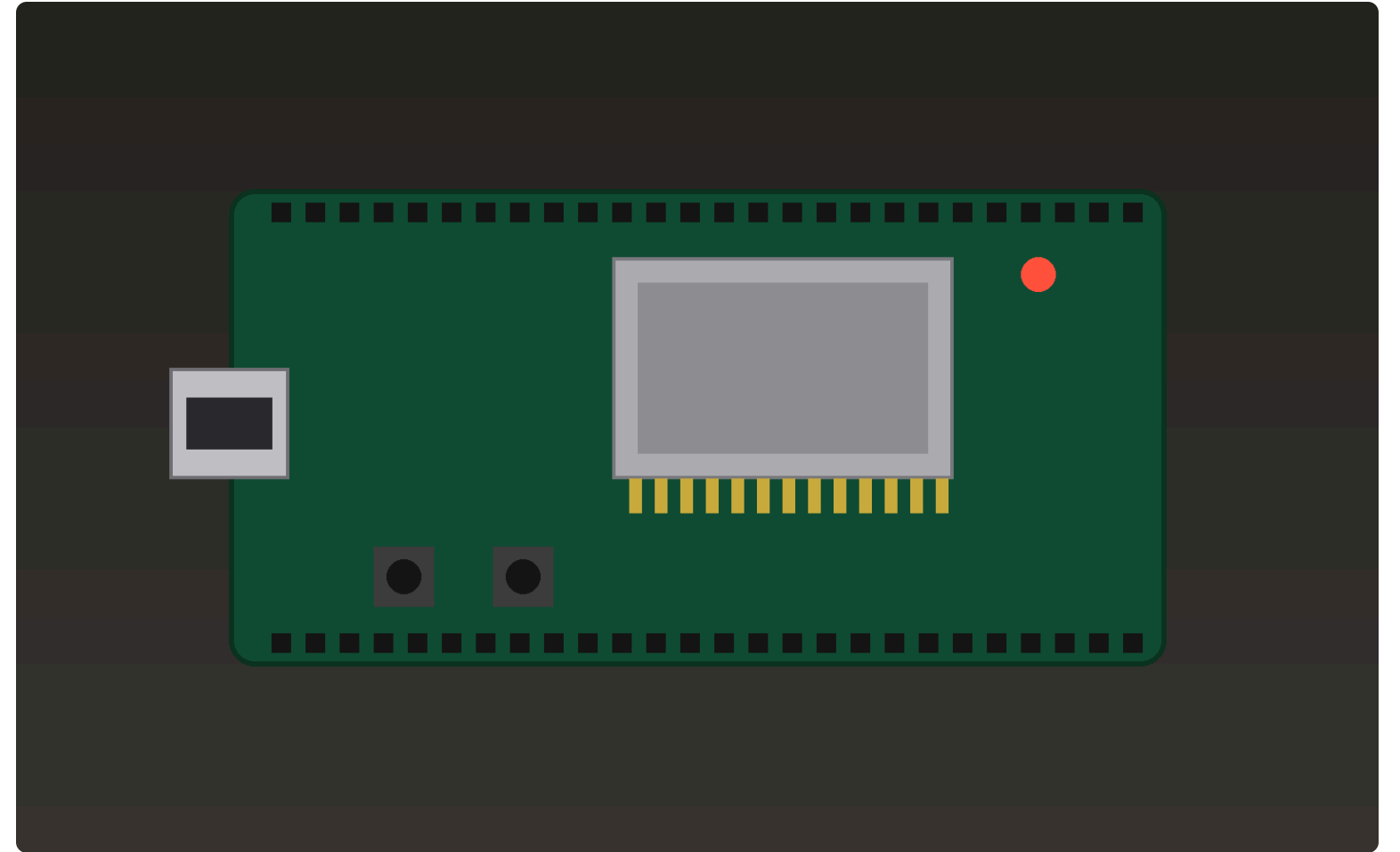
# MANUAL OR WITH AN AGENT?

| Task         | Manual            | With agent  |
|--------------|-------------------|-------------|
| Flashing     | 1 min             | automatic   |
| Reading logs | eyes on screen    | continuous  |
| Testing      | when you remember | every build |



# SETUP

- Laptop with agent
- USB hub
- Test board



# Thank you!

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Questions? Comments? Let me know.